

Shravya Munugala

Providence, RI | +1 (401) 215-3306 | shravya_munugala@brown.edu | [Portfolio](#) | [GitHub](#) | [LinkedIn](#)

EDUCATION

Brown University — Master of Science, Computer Science (GPA: 3.75/4.0)

Providence, RI | Expected May 2027

Coursework: Data Engineering, Advanced Deep Learning, Natural Language Processing, Scientific Visualization

Harvard University — Cross-registered Student

Cambridge, MA | Sep 2026 – Dec 2026

Coursework: AC215 MLOps & LLMops: Production AI Systems

Vellore Institute of Technology — Bachelor of Technology, Computer Science (GPA: 3.97/4.0)

Chennai, India | May 2025

Coursework: Data Structures & Algorithms, Operating Systems, Database Systems, Distributed Systems, Object-Oriented Programming

EXPERIENCE

Graduate Research Assistant | **Brown University**

Feb 2026 – Present

- Built a statewide Python/Selenium crawler that models menus as navigation graphs and replays click paths in headless Chrome, testing whether families can reach critical pages on 395 Massachusetts district websites
- Located special education pages for 98.5% of districts, matching human coders on 83% of choices and saving 50+ hours of manual review per cycle
- Engineered fault tolerance for 10-hour batch runs with checkpointed resume, retries, dead-domain fallback, and API failover, completing 12 consecutive statewide runs with zero lost progress
- Built a searchable data explorer in Streamlit for 6 non-technical researchers, and reviewed a co-developer's code to catch regressions

Software Engineering Intern | **Kilowant**

May 2026 – Aug 2026

- Built a full-stack Angular/TypeScript and Node.js app, adopted company-wide, replacing a legacy admin dashboard and unifying buyer, supplier, contract, and utility data from a 1.5 GB SQL dataset, cutting account lookups from ~5 min to under 10 s
- Decoupled data processing from the API layer using typed TypeScript models and the repository pattern, enabling migration from static snapshots to live SQL Server-backed APIs and cutting new-feature integration time by ~40%
- Built a Python data pipeline that crawled 17 sources in parallel with retries, validation, and deduplication, feeding an LLM workflow that generated personalized emails for 7,000+ leads at a 32% open rate

Cloud Computing Intern | **Pentabrain Solutions Pvt Ltd**

Apr 2025 – Jul 2025

- Developed serverless REST endpoints with AWS Lambda and API Gateway for client onboarding and billing workflows
- Contributed to a GitHub Actions CI/CD pipeline automating backend build and deployment, cutting release time by 60%

PROJECTS

AgentRewind (llm-run-recorder on PyPI) | *Python, SQLite, PostgreSQL, GitHub Actions*

Jun 2026 – Present

- Published an open-source Python package that records LLM agent runs (tool calls, latency, tokens, execution spans) to SQLite and replays them locally with zero provider API calls, replaying a 50-call run in under 2 s vs. ~90 s live
- Built OpenAI- and Anthropic-compatible instrumentation and structural span-tree diffing to pinpoint where two agent runs diverge
- Made async tracing concurrency-safe across 12 parallel runs; shipped 4 releases via GitHub Actions CI/CD
- Built a pluggable storage layer with an optional PostgreSQL backend; one 145-test suite covers both backends at 99.6% coverage

DriftGuard – RAG Quality Monitoring Platform | *Python, FastAPI, FAISS, Redis, Docker, Prometheus*

Jan 2026 – Present

- Built a FastAPI/FAISS vector search service over 13,698 token-bounded chunks, with zero-downtime re-indexing via atomic swaps
- Designed a statistical drift detector using PCA and two-sample KS tests, catching 20/20 topic shifts in a median of 50 queries
- Eliminated false alarms, from 9 per 200 windows to 0, by adding three-window hysteresis to Prometheus alerts
- Added Redis query caching and shared drift state across 3 replicas, cutting p50 query latency from 10.9 ms to 1.3 ms

Elyssa – Patented AI Mental Health Assistant | *Python, FastAPI, Docker, Kubernetes*

Feb 2025 – Jul 2025

- Architected a scalable five-service Python/FastAPI microservices system on Kubernetes for vision inference, sentiment analysis, and response generation, sustaining 200 req/s at p95 < 300 ms
- Added circuit breakers, timeouts, readiness probes, and 2–8 replica autoscaling, absorbing a 5x load spike with zero dropped requests
- Validated failure and recovery paths with 157 tests at 97% branch coverage, including retry and circuit-breaker behavior

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, Java, C, C++, SQL

Frameworks & Libraries: Angular, Node.js, FastAPI, React, Selenium, Playwright, mypy, Pandas, NumPy, PyTorch

Cloud & DevOps: AWS (Lambda, API Gateway, S3), GCP, Docker, Kubernetes, GitHub Actions, CI/CD, Prometheus, Git, Linux

Databases & Testing: SQL Server, SQLite, PostgreSQL, Redis, pytest, Unit Testing

Core Concepts: Distributed Systems, System Design, REST APIs, Scalability, Microservices, Fault Tolerance

AI/ML: Machine Learning, NLP, RAG, LLMs, Hugging Face, Embeddings, Vector Search (FAISS)

Certifications: AWS Certified Cloud Practitioner, Google Cloud Digital Leader

PATENTS & PUBLICATIONS

- Interactive System for Emotional Well-Being and Assisted Care, Indian Patent App. No. 202541093582A (published), 2025
- Alzheimer's Disease Classification Using CUDA-Accelerated DenseNet201 and VGG16, IEEE ICICNIS 2024

LEADERSHIP & ACTIVITIES

Writing Associate | **Sheridan Center, Brown:** led 150+ one-on-one coaching sessions; ELP associate

Nov 2025 – Present

Head of HR & Management | **PlaceXP Club, VIT Chennai:** led 7+ events reaching 200+ students

2024 – 2025